



AUTOSAR Adaptive - Enabler for High Performance Connectivity and Automated Driving

2017-11-07 Dr. Günther Heling

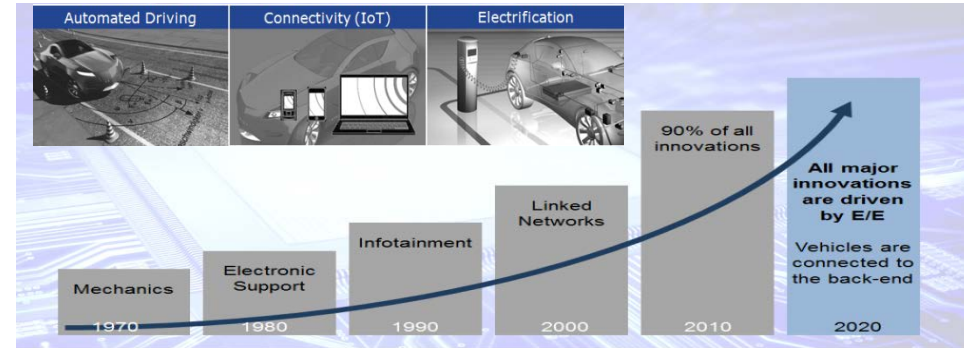
2002 – 2017

„Perfection of Automotive Embedded Systems“

The Challenge 15 Years ago

1. Vehicles need to offer a tremendous increase of **software based functionality**

→ *We have to do much more*



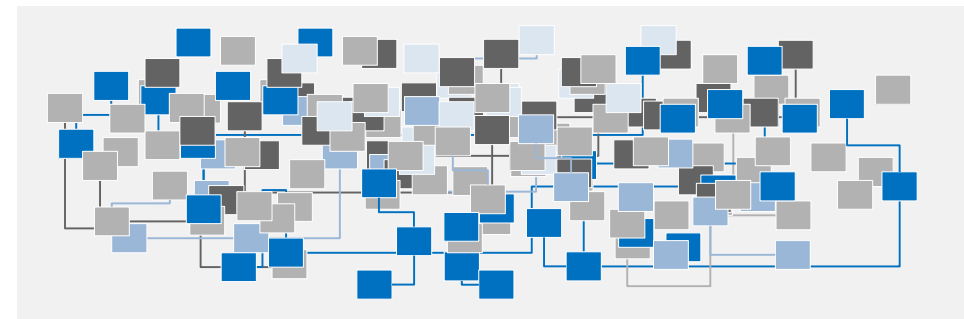
2. **Time to market** must be shorter and budgets must be smaller

→ *We have to get it done faster*



3. **E/E complexity** and interconnection between functions is increasing

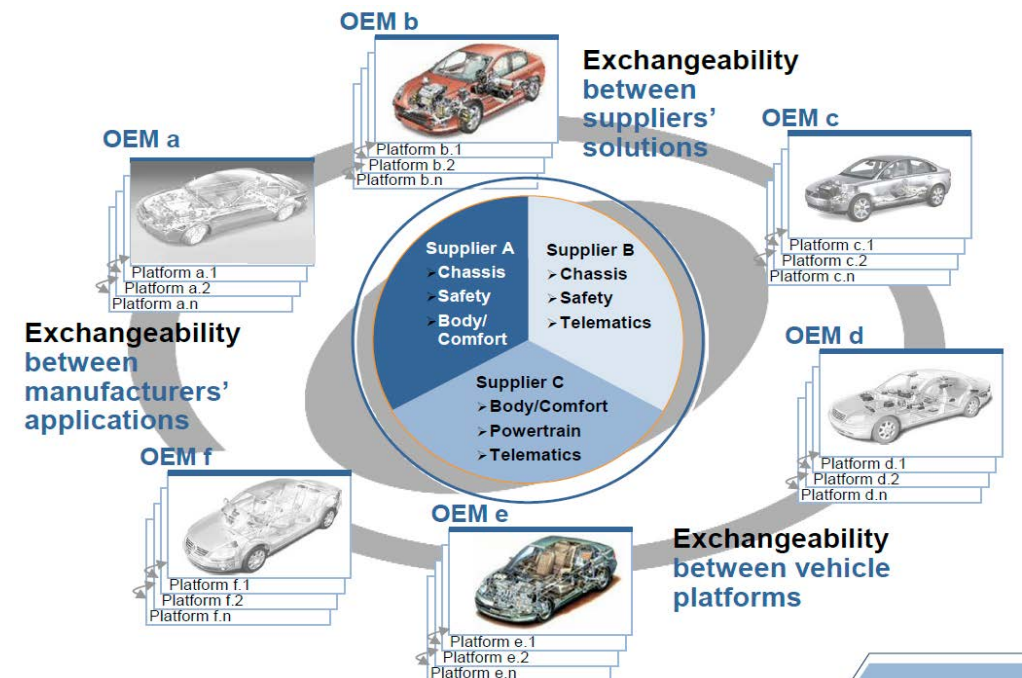
→ *We have to master it*



R e u s e !

Vision:

- ▶ **We all speak the same language:**
 - ▶ we use the same terms
 - ▶ we use the same powerful methods
 - ▶ we use standardized data exchange formats
- ▶ **One solution for one problem**



Source: AUTOSAR.org

AUTOSAR Classic - **many variants** for many OEMs

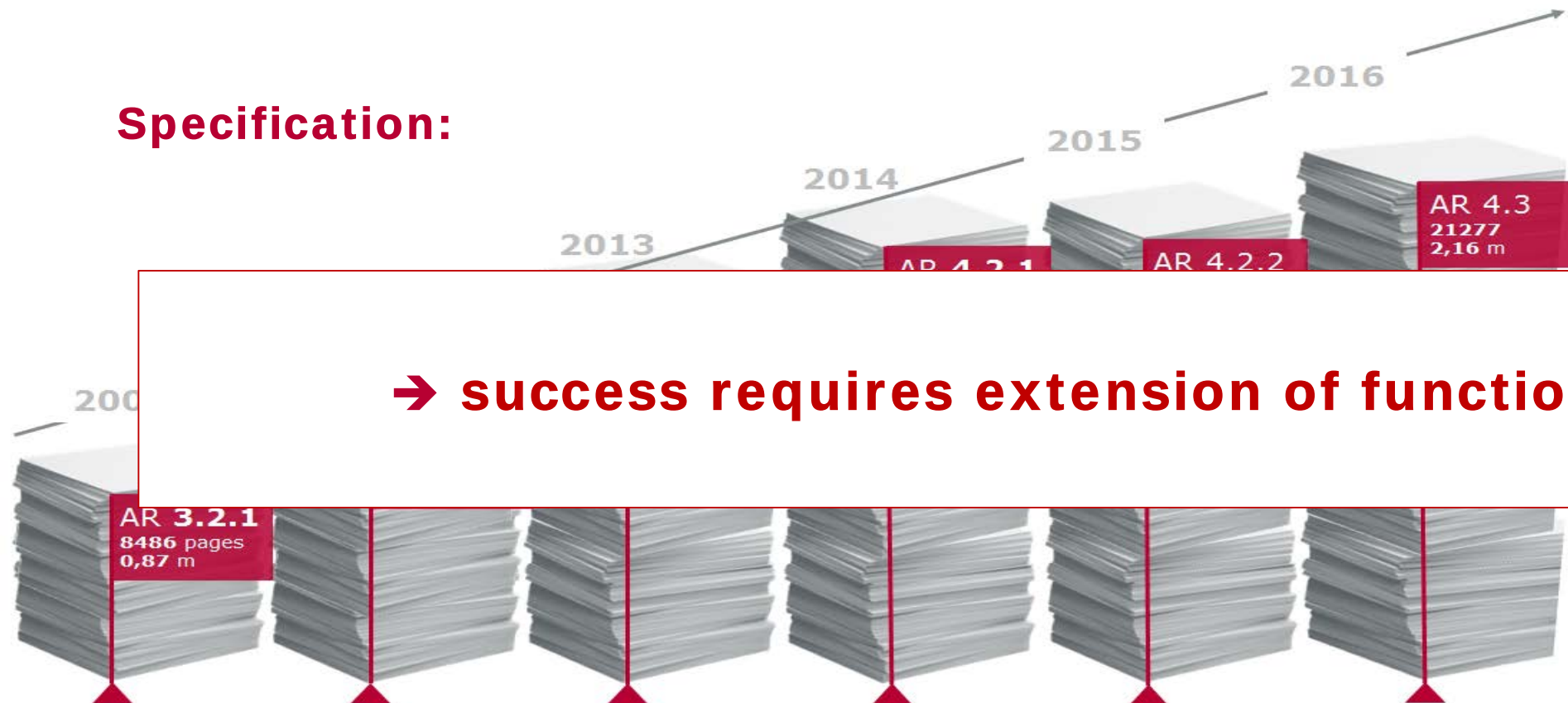
Program	AUTOSAR version	Configuration	Bus System
1	3.2.1	Legacy LDF + DBC + Fibex	CAN, LIN, FlexRay
2	3.2.2	AUTOSAR SD	CAN, LIN, FlexRay
3	3.2.x	Legacy LDF + DBC	CAN, LIN
4	3.2.x	Legacy Fibex	CAN, LIN, FlexRay
5	3.2.x	Legacy LDF + DBC	CAN, LIN
6	3.x	Legacy LDF, DBC + Fibex	CAN, LIN, FlexRay
7	3.x	Legacy LDF, Fibex	CAN, LIN, FlexRay

→ success requires adaptation to specific needs

15	4.0.3, 4.2.1 (Eth)	AUTOSAR SD	CAN, LIN, Eth
16	4.2.1	SD 4.2.1	CAN, CAN-FD, LIN, FlexRay, Eth
17	4.2.2	AUTOSAR SD + DEXT	CAN, LIN, FlexRay, Eth
18	4.2.2	AUTOSAR SD	CAN, LIN
19	4.2.2	Legacy LDF + DBC	CAN, LIN, Eth, J1939
20	4.2.2 + 4.3 features	AUTOSAR SD + DEXT	CAN, CAN-FD, FlexRay, Eth, LIN
21	4.2.2 + 4.3 features	AUTOSAR SD + DEXT	CAN, CAN-FD, FlexRay, Eth
22	4.2.x	Legacy LDF + DBC	CAN, LIN, Eth
23	4.X	Legacy LDF + DBC	CAN, LIN
24	4.X	Legacy LDF + DBC	CAN, LIN, Eth
25	4.X	Legacy LDF + DBC	CAN, LIN
26	4.X	Legacy LDF + DBC	CAN, LIN
27	4.x	Legacy LDF + DBC , ARXML	CAN, LIN

AUTOSAR Classic – **many versions** with growing functionality

Specification:



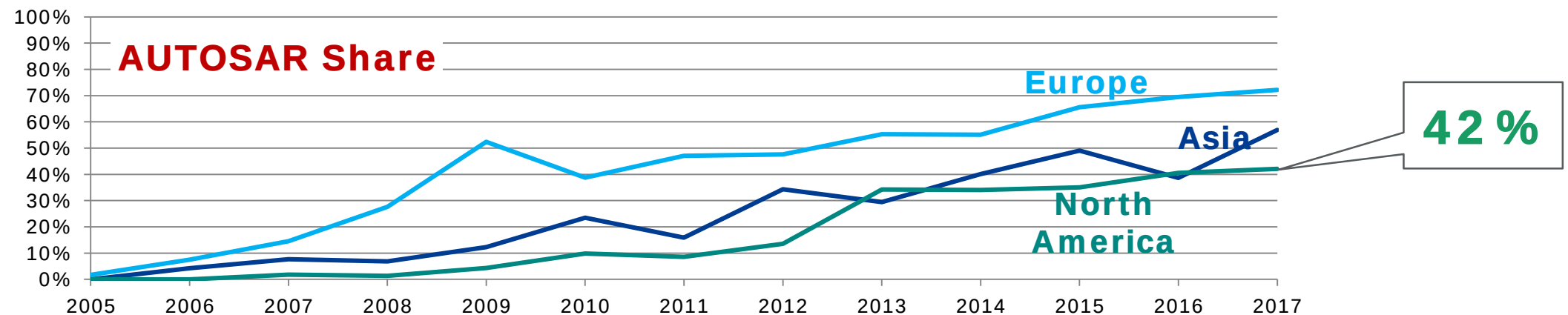
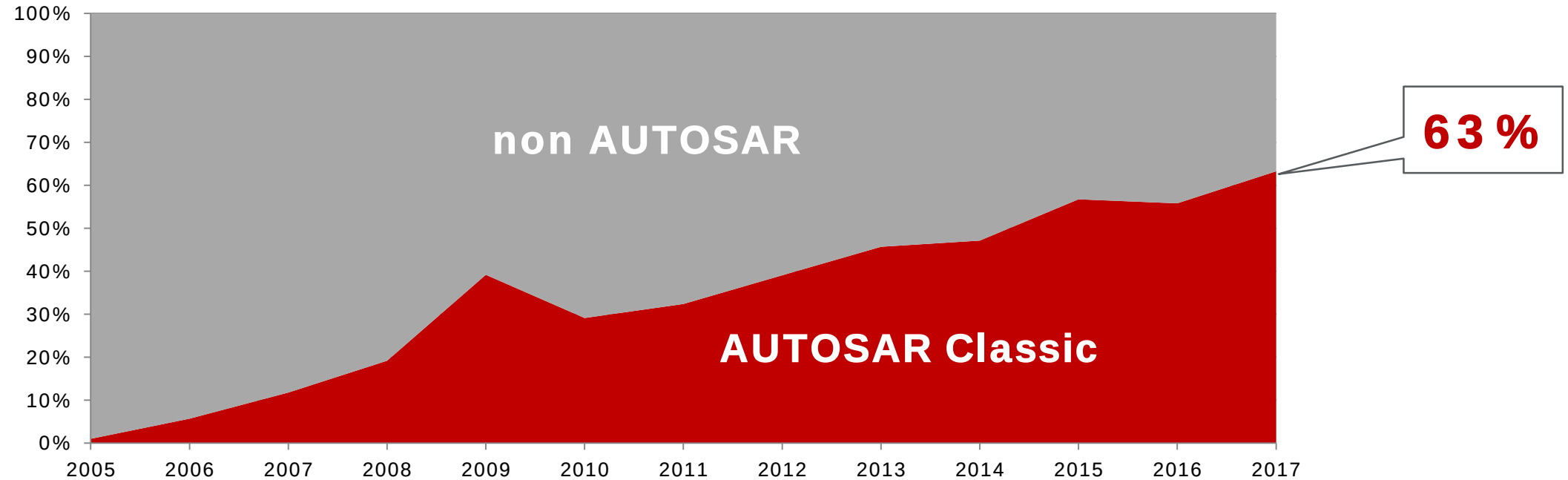
Changes in 4.3

- ▶ **Ethernet**
 - traffic shaping
 - queue optimization
 - SOMEIP-TP
- ▶ **Security**
 - new crypto interfaces
 - handling
 - stics

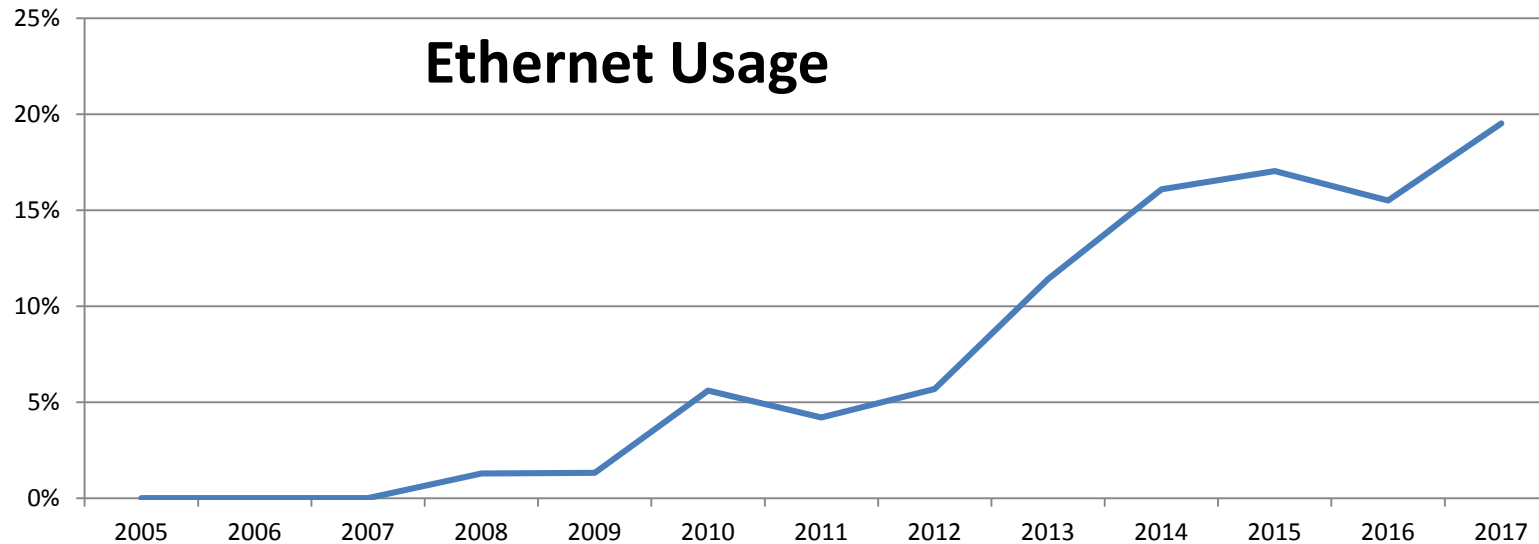
Implementation:

	AUTOSAR 3	AUTOSAR 4
Number of BSW Modules	68	104
Lines of Code	100%	271%

AUTOSAR grows continuously – mission accomplished



Communication changes – Ethernet & Services



Source: Vector's evaluation
based on number of projects

- Signal based communication is supplemented by **Service oriented communication**

Signal based	Service oriented
signals	complex data
binding at compile time	binding at run time (service discovery)
static closed system	dynamic open system

2017 – 2035

„Automotive goes IT“

Automotive Mega Trends

2030*

Level 5



Electrification

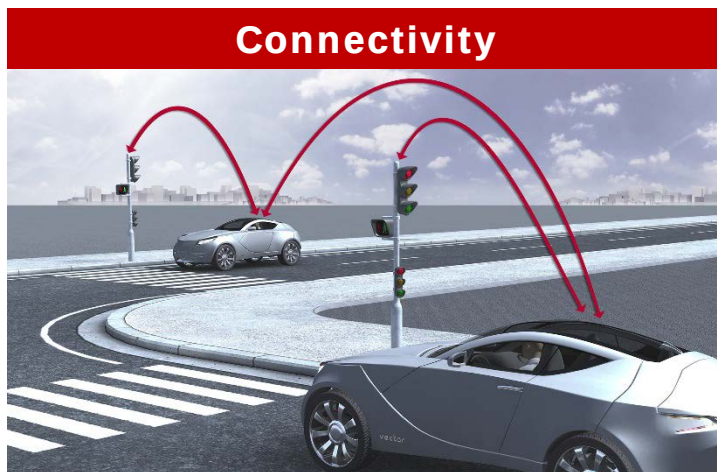


2035*

**Production
EV > CbV**

need

Connectivity



Offboard Eco Systems



E/E changes

Technology Drivers

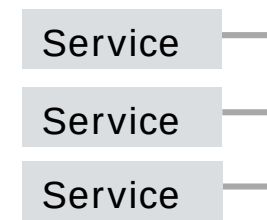
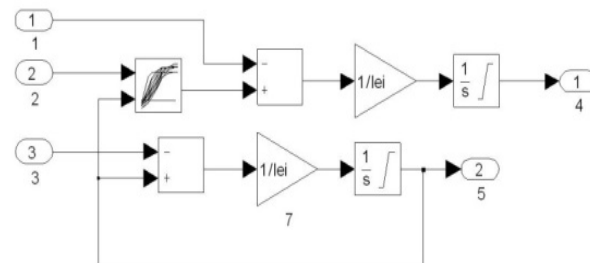
Automated Driving
Electrification
Connectivity
Offboard Eco Systems

Implications

now

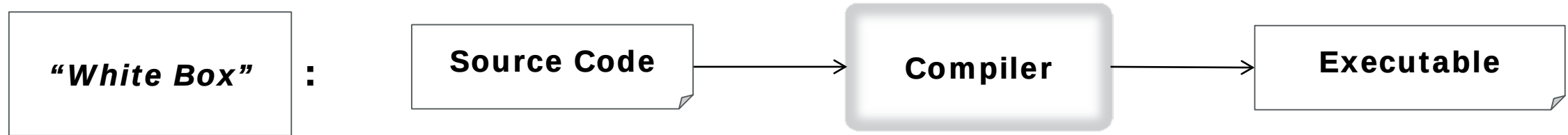
in future (additionally)

Platform	AUTOSAR / OSEK	AUTOSAR Adaptive, Linux, ...
Communication	Signals	Services with complex data
Configuration	Static	Dynamic
Architecture	ECU / Component	Service
Development Methods	Deterministic	AI and Agile

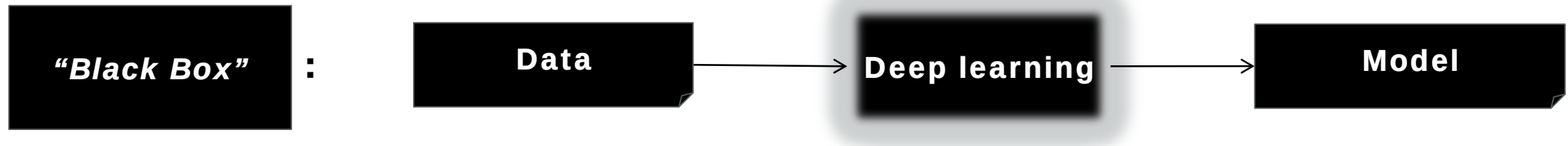


Development Methods and Tools

1. Deterministic algorithms ...



... are supplemented by **Artificial Intelligence (AI)**



2. Agile Development – to be tailored for automotive needs

Challenges:

- ▶ Scalable agile Development ("SCRUM of SCRUMs")
- ▶ Development according ISO 26262

The Vision of AUTOSAR Adaptive

R e u s e !
from other domains

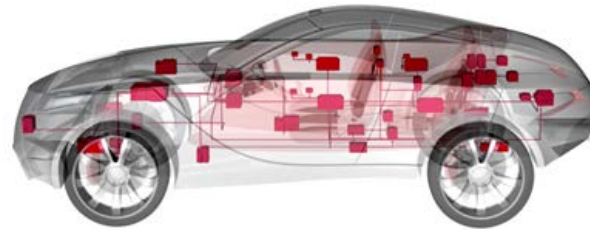
Adaptive – best of two worlds

Cloud / Backend



- ▶ support of high performance processors
- ▶ high bandwidth
- ▶ service based architectures
- ▶ open source
- ▶ dynamic and updatable
- ▶ internet

Embedded Systems



AUTOSAR Classic

- ▶ safe
- ▶ secure
- ▶ embedded integration and debugging
- ▶ automotive supply chain
- ▶ automotive communication protocols
- ▶ automotive diagnostics

Adaptive – best of two worlds

Cloud / Backend

- system and mobility strategies
- deep learning

On board HP Computing

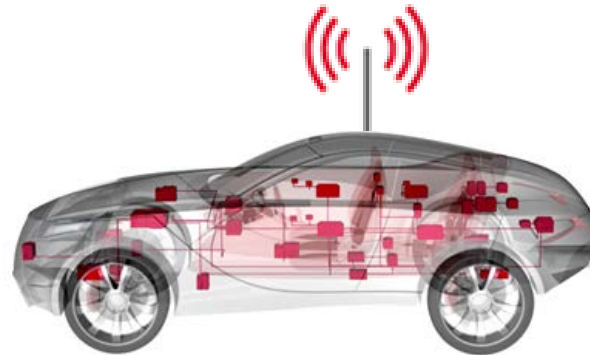
- multipurpose computing servers
- connectivity, gateways, HMI
- automated driving
- ! mastered by OEM

AUTOSAR Adaptive

Embedded Systems

- intelligent sensors and actuators
- basic functions
- fallback computing

AUTOSAR Classic



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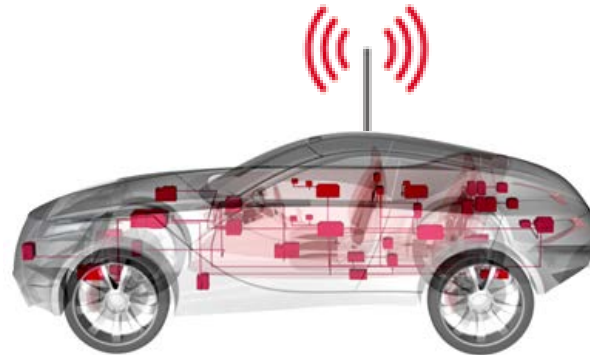
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! **dynamic and updatable**

! **internet**

! **safe**

! **secure**

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There are more reasons to use Adaptive !

Let's have a look at the development process:

Classic AUTOSAR

- ▶ static
- ▶ monolithic / top-down
- ▶ model-based
- ▶ configuration at compile time

- ➔ maximal efficiency
- ➔ quick local changes not supported

AUTOSAR Adaptive

- ▶ dynamic
- ▶ modular / parallel
- ▶ model-based and software-based
- ▶ configuration at runtime

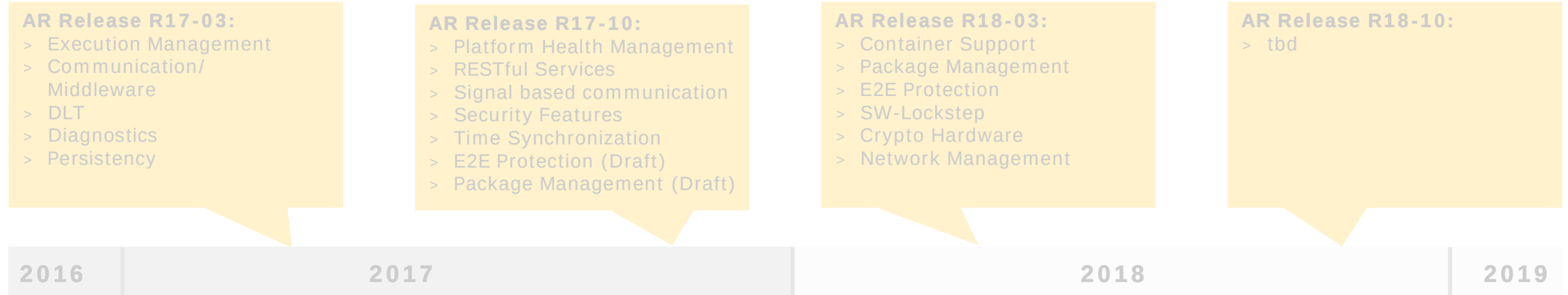
- ➔ maximal flexibility
- ➔ quick local changes well supported

- ▶ Good software design is more important than small footprint
- ▶ Software is developed by computer scientist

- ➔ good fit to AUTOSAR Adaptive
- ➔ good fit to AUTOSAR Adaptive

AUTOSAR Adaptive may be the right choice even for smaller ECUs

Now it's time to get on board



1. let's define **one** powerful standard → **Contribute to the specification**
2. gain experience early → **Start evaluation projects now**
3. combination of automotive embedded & IT is challenging → **Care for skilled staff**

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